Mid-term Project Report: Privacy-Preserving Sealed-Bid Auctions

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1 System Description

A classical sealed-bid auction requires a trusted auctioneer to open every bid and declare a winner, which creates a fundamental privacy problem: the auctioneer learns each bidder's private valuation and can exploit it in ongoing negotiations or leak it to favoured participants. Even an honest auctioneer is a single point of failure whose compromise exposes every participant at once. The question that motivates this project is whether the same auction outcome can be computed without requiring any single party to see the individual bids at all.

We are building a **privacy-preserving sealed-bid auction** in which no party — including the servers running the auction — ever sees any individual bid in the clear. The system follows the secret-sharing multiparty computation approach of Bogetoft et al., in which each bid is split into shares distributed across a small set of computing servers. The servers jointly compute the auction outcome directly on these shares, and only the final result is reconstructed in the clear. This generalises the trust assumption of a classical auction: rather than relying on a single auctioneer to behave honestly, privacy holds as long as no sufficient threshold of servers collude.

The core computational task is finding the maximum of the secret-shared bids, for which we will implement the **bit-slice maximum circuit** of Kurosawa and Ogata (2002). This circuit identifies the winning bid by scanning all bids simultaneously one bit position at a time, without ever reconstructing any individual bid during the computation. A second scan, with the winner's contribution removed obliviously inside the computation, yields the second-highest bid without revealing the winner's identity mid-protocol.

The system will support two auction mechanisms:

- **First-price auction:** the highest bidder wins and pays their own bid, determined by the first scan.
- **Second-price (Vickrey) auction:** the highest bidder wins but pays the second-highest bid, determined by the oblivious second scan. This incentivises truthful bidding, since submitting one's true valuation is a dominant strategy.

The threat model targets the **semi-honest, honest-majority** adversary: computing servers follow the protocol correctly but may attempt to infer bid values from their local views, with privacy holding as long as no threshold of servers collude. The protocol reveals only the winner's identity and the clearing price — not the rank, ordering, or approximate value of any losing bid.

2 Planned Individual Contributions

The implementation is divided into four components, each owned by one group member: the secure computation environment and testing infrastructure, the bit-slice maximum circuit, the Vickrey extension and performance evaluation, and the bid input and output reconstruction pipeline.

2.1 Aarav Oswal (230012) — Computation Environment and Testing

Aarav will set up the **secure multiparty computation environment** and build the **correctness testing infrastructure**. This involves configuring the multi-server environment for the semi-honest, honest-majority setting and verifying that it produces correct outputs on synthetic inputs before any auction-specific computation is introduced. He will additionally develop a test suite that validates the auction circuit's output against a plaintext reference implementation across a range of inputs, including edge cases such as ties, a single bidder, and all-equal bids. This test suite will serve as the primary validation mechanism for all components as they come together into the final system.

2.2 Ch. Rishitha (230311) — Bit-Slice Maximum Circuit

Rishitha will implement the **bit-slice maximum circuit** of Kurosawa and Ogata. The circuit identifies the maximum among n secret-shared bids by scanning their bits from most to least significant, narrowing the set of surviving candidates at each step without reconstructing any bid in the clear. The total cost is $O(nk)$ secure operations for n bidders with k-bit bids, linear in both the number of bidders and the bid length. The output of this circuit — the index and value of the winning bid — forms the **first-price auction core** and serves as the starting point for the Vickrey extension.

2.3 R. Charan (230819) — Vickrey Extension and Performance Evaluation

Charan will extend the first-price circuit to support the **Vickrey mechanism** and evaluate the performance of the full system. The Vickrey extension adds a second scan over the secret-shared bids in which the winner's contribution is removed obliviously, so that the second-highest bid is returned without revealing the winner's identity mid-protocol. The asymptotic cost of the two scans combined remains linear. He will additionally measure the performance of the combined system across varying numbers of bidders and bid lengths, characterising how the implementation scales in practice.

2.4 Poorvie Sadagopan (230759) — Input and Output Pipeline

Poorvie will implement the **input and output layers** of the system. On the input side, this involves encoding bidder inputs as secret shares and distributing them to the computing servers such that no server can reconstruct any bid individually. On the output side, she will implement the reconstruction step that reveals only the winner's identity and clearing price to the relevant parties, enforcing the privacy guarantee at the boundary where the final result re-enters the clear. The correctness of both layers will be verified against the privacy target of Section 1: that the protocol reveals the winner and clearing price, and nothing else.